

HII: Ionized Hydrogen H^+

• $H^+ + e^-$

• photons from stars

$$\langle E \rangle = 13.6 \text{ eV} = kT$$

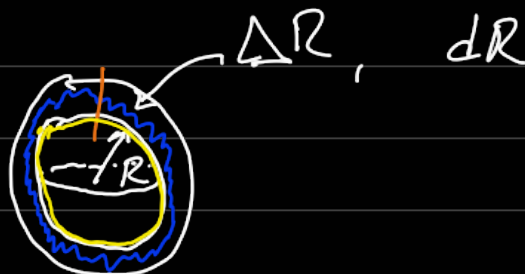
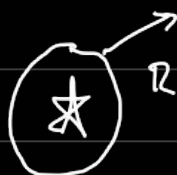
• reattachment: $H^+ + e^- = H + \gamma$



• Ionization: $S_\star$: (# photons/time)

$$N_\gamma = S_\star t$$

• Recombination: $R = \alpha n_p$



• i) Set up above:

look n_H within ΔR

$$n_H : \frac{N_\gamma}{\Delta V}$$

$$(SA) = 4\pi R^2$$

$$\Delta V = 4\pi R^2 \Delta R$$

$$N_\gamma = n_H 4\pi R^2 \Delta R = S_\star \Delta t \quad *$$

$$\frac{\Delta R}{\Delta t} (n_H 4\pi R^2) = S_\star \quad \left| \quad \frac{dR}{dt} (n_H 4\pi R^2) = S_\star \right.$$

take integral

$$\int n_H 4\pi R^2 dR = \int S_\star dt \quad *$$

$$R_I^3 \frac{4}{3} \pi n_H = S_* t$$

$$R_I \propto t^{1/3} \quad (\text{Ionization})$$

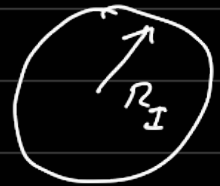
$$* R = \alpha n_p \quad (\text{per proton})$$

• recombinations / sec

→ Total recombinations in ionization region

$$N_c = n_e V \leftarrow \frac{4}{3} \pi R_I^3$$

$$N_c R = R_{\text{tot}}$$



$$R_{\text{tot}} = n_e \frac{4}{3} \pi R_I^3 \propto n_p$$

$$n_H = n_e = n_p$$

$$\hookrightarrow n_H^2 \frac{4}{3} \pi R_I^3 \propto$$

* Strömgen radius:

$$\text{Defined by } R_{\text{tot}} = S_*$$

$$S_* = n_e \frac{4}{3} \pi R_s^3 \propto n_p$$

$$R_s = \left(\frac{3 S_*}{n_e n_p \alpha 4 \pi} \right)^{1/3} *$$

* Recall:

$$N_\gamma = S_* \Delta t = 4 \pi R^2 \Delta R n_H$$

$$\text{limit: } \int S dt = \int 4 \pi R^2 \left(n_0 \left(\frac{R}{R_0} \right)^{-2/3} \right) dR$$

$$* \left[\frac{Fe}{H} \right] = \log_{10} \left(\frac{Z}{Z_0} \right) \quad \left| \quad \begin{array}{l} Z = \left(\frac{Fe}{H} \right) \\ Z_0 = \left(\frac{Fe}{H} \right)_0 \end{array} \right. *$$

$$\underline{Z_0 = .62}$$

- stellar halo: up to $\sim \frac{1}{300} Z_0$

* Initial Mass function: (IMF) $\xi(M + \Delta M)$

- $\xi(M) \Delta M$: # stars born between mass M
Volume

$$\xi(M) = \xi_0 \left(\frac{M}{M_0} \right)^{-\alpha}$$

α : varies within Milky way

$\alpha \approx 2.35$

 $\alpha \approx 1.7$

$$\int_{M_A}^{M_B} \xi(M) dM = \text{# of stars w masses between } M_A \text{ and } M_B$$

* Mass: $\int_{M_A}^{M_B} M \cdot \xi(M) dM$: Mass of stars with mass between M_A and M_B

$$M \xi(M) dM$$

PS 31 Total $M_A = 0.5$
 $M_B = 150$

$$M_{\text{tot}} = \int_{0.5}^{150} M \xi(M) dM$$

fraction: $\frac{\int_{0.5}^{150} M \xi(M) dM}{M_{\text{tot}}}$

Optional | Mean stellar mass:

• Distribution: $f(x)$ $\int f(x) dx = 1$

$$\langle E \rangle = \int E f(x) dx$$

$M \xi(M) \leftarrow$ Mass distribution

$$\langle M \rangle = \int \underbrace{M}_{\xi} (M \xi(M)) dM$$

